

Short Answer Questions

1. What is jQuery?

Ans: **jQuery** is a fast, lightweight **JavaScript library** that simplifies HTML document traversal, DOM manipulation, event handling, animation, and AJAX interactions for web development.

2. List the features of jQuery.

Ans: Main features of jQuery include:

- Easy DOM manipulation
- Cross-browser compatibility
- Simplified event handling
- Built-in animation and effects
- AJAX support
- Lightweight and fast
- Chainable methods for efficient coding

3. Write the syntax of jQuery.

Ans: Basic syntax:

`$(selector).action();`

Where:

- `$` → jQuery symbol
- `selector` → selects HTML element(s)
- `action()` → operation performed on selected element

Example:

`$("#p").hide();`

4. What is the purpose of `$(document).ready()` function?

Ans: The `$(document).ready()` function ensures that the **JavaScript code runs only after the HTML document has completely loaded**.

Example:

```
$(document).ready(function(){  
    $("#p").click(function(){  
        alert("Paragraph clicked");  
    });  
});
```

5. Explain jQuery selectors.

Ans: **jQuery selectors** are used to **select and manipulate HTML elements** based on their name, ID, class, or attributes.

Common selectors include:

Selector	Example	Description
Element	<code>\$("#p")</code>	Selects all <code><p></code> elements
ID	<code>\$("#id")</code>	Selects element with specific ID
Class	<code>\$(".class")</code>	Selects elements with a class
Universal	<code>\$("#*")</code>	Selects all elements

6. What is an ID selector in jQuery?

Ans: An **ID selector** selects a single HTML element using its **ID attribute**.

Syntax:

```
$("#idname")
```

Example:

```
$("#title").hide();
```

This hides the element with ID **title**.

7. What is a class selector in jQuery?

Ans: A **class selector** selects all elements that have a specific **class attribute**.

Syntax:

```
$(".classname")
```

Example:

```
$(".demo").css("color","red");
```

This changes the text color of all elements with class **demo**.

8. What are jQuery methods?

Ans: **jQuery methods** are built-in functions used to **perform actions on selected HTML elements**.

Examples:

- hide()
- show()
- css()
- html()
- append()
- remove()

Example:

```
$("#p").hide();
```

9. Explain DOM manipulation in jQuery.

Ans: **DOM manipulation** refers to modifying the **structure, content, or style of HTML elements dynamically** using jQuery.

Common DOM manipulation methods:

- append() → adds content at the end
- prepend() → adds content at the beginning
- remove() → removes elements
- empty() → removes content

Example:

```
$("#p").append(" Welcome!");
```

10. What is a callback function in jQuery?

Ans: A callback function is a function that is executed after another function completes its execution.

Example:

```
$("#p").hide("slow", function(){  
    alert("Paragraph is hidden");  
});
```

Here the alert message appears after the hide animation finishes.

11. List different jQuery events.

Ans: Common jQuery events include:

- click()

- dblclick()
- mouseenter()
- mouseleave()
- hover()
- keypress()
- keydown()
- submit()
- focus()
- blur()

12. What are jQuery effects?

Ans: **jQuery effects** are built-in functions used to **create animations and visual effects** on web pages.

They help in making webpages **interactive and dynamic**.

Examples include hiding, showing, fading, and sliding elements.

13. Write any four jQuery effects methods.

Ans: Four common jQuery effects methods are:

- hide()
- show()
- fadeIn()
- fadeOut()
- slideUp()
- slideDown()

Example:

```
$("#p").fadeOut();
```

14. What is AJAX?

Ans: **AJAX (Asynchronous JavaScript and XML)** is a technique used to **send and receive data from a server without reloading the entire webpage**.

It helps create **dynamic and faster web applications**.

15. List AJAX methods available in jQuery.

Ans: Common jQuery AJAX methods include:

- load() – loads data from server
- get() – requests data using HTTP GET
- post() – sends data using HTTP POST
- ajax() – performs advanced AJAX requests
- getJSON() – retrieves JSON data

Long Answer Questions

1. Explain jQuery and its advantages in web development

Ans: **jQuery** is a fast, lightweight, and feature-rich **JavaScript library** designed to simplify client-side scripting of HTML. It makes tasks like **HTML document traversal, DOM manipulation, event handling, animation, and AJAX interactions** much easier with simple and concise syntax. jQuery uses the **\$ symbol** and provides built-in functions that reduce the amount of JavaScript code needed.

Example

```
$("#p").hide();
```

This single line hides all paragraph elements on a webpage.

Advantages of jQuery

1. **Simplified JavaScript Programming**
Complex JavaScript tasks can be written with fewer lines of code.
2. **Cross-Browser Compatibility**
jQuery automatically handles browser compatibility issues.
3. **Easy DOM Manipulation**
It allows easy modification of HTML elements and attributes.
4. **Powerful Event Handling**
User interactions like clicks and mouse movements can be easily handled.
5. **Built-in Animation Effects**
jQuery provides animation effects such as fade and slide.
6. **AJAX Support**
Allows loading data from the server without refreshing the webpage.
7. **Lightweight and Fast**
The library is small in size and loads quickly.

2. Describe how to load and use jQuery in a webpage with example

Ans: jQuery must be **included in the HTML page** before using it. This can be done in two ways.

1. Using CDN (Content Delivery Network)

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

2. Downloading the jQuery File

Download jQuery and include it locally.

```
<script src="jquery.js"></script>
```

Example Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>jQuery Example</title>
```

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

```
<script>
```

```
$(document).ready(function(){
    $("button").click(function(){
        $("p").hide();
    });
});
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<p>This is a paragraph.</p>
```

```
<button>Hide Paragraph</button>
```

```
</body>
```

```
</html>
```

Explanation

- `$(document).ready()` ensures the page loads before executing code.
- When the button is clicked, the paragraph is hidden.

3. Explain jQuery selectors with suitable examples

Ans: **jQuery selectors** are used to select HTML elements so that actions can be performed on them.

Types of Selectors

1. Element Selector

Selects elements by tag name.

```
$("p").hide();
```

Hides all paragraph elements.

2. ID Selector

Selects an element using its ID.

```
$("#header").css("color", "blue");
```

Changes the text color of element with ID **header**.

3. Class Selector

Selects elements with a specific class.

```
$(".demo").hide();
```

Hides all elements with class **demo**.

4. Universal Selector

Selects all elements.

```
$("*").hide();
```

Example HTML

```
<p id="p1">Paragraph 1</p>
```

```
<p class="demo">Paragraph 2</p>
```

4. Discuss different jQuery methods used for DOM manipulation

Ans: **DOM manipulation** means changing the structure or content of HTML elements dynamically.

Common DOM Manipulation Methods

Method	Description
append()	Adds content at the end
prepend()	Adds content at the beginning
before()	Inserts content before element
after()	Inserts content after element
remove()	Removes element
empty()	Removes all child elements

Example

```
$("p").append(" Welcome!");
```

Adds the text **Welcome!** to the paragraph.

Another example:

```
$("#div").remove();
```

Removes all div elements.

5. Explain callback functions in jQuery with example

Ans: A **callback function** is a function that executes **after another function completes its execution**.

This ensures that certain code runs only after an operation finishes.

Example

```
$("p").hide("slow", function() {  
    alert("Paragraph hidden successfully");  
});
```

Explanation

1. hide("slow") hides the paragraph slowly.

2. After the animation finishes, the callback function executes.
3. The alert message is displayed.

6. Explain various jQuery events with examples

Ans: **Events** are actions performed by users such as clicking, typing, or moving the mouse.

jQuery provides methods to handle these events easily.

Common jQuery Events

Event	Description
click()	Mouse click
dblclick()	Double click
mouseenter()	Mouse enters element
mouseleave()	Mouse leaves element
keypress()	Keyboard key pressed
submit()	Form submission

Example

```
$("#button").click(function() {
    alert("Button clicked");
});
```

When the button is clicked, an alert appears.

7. Describe jQuery effects such as hide, show, fade, and slide with examples

Ans: jQuery provides built-in **animation effects** to make webpages interactive.

1. hide()

Hides an element.

```
$("#p").hide();
```

2. show()

Displays a hidden element.

```
$("#p").show();
```

3. fadeIn() and fadeOut()

Creates fading animation.

```
$("#p").fadeOut();
```

4. slideUp() and slideDown()

Creates sliding animation.

```
$("#p").slideDown();
```

Example Program

```
$("#button").click(function() {
    $("#p").slideUp();
});
```

8. Explain the concept of AJAX in jQuery

Ans: **AJAX (Asynchronous JavaScript and XML)** is used to **exchange data with a server without refreshing the entire webpage.**

jQuery simplifies AJAX operations.

AJAX Methods

Method	Description
load()	Loads data from server
get()	Sends HTTP GET request
post()	Sends HTTP POST request
ajax()	Advanced AJAX method

Example

```
$("#content").load("data.txt");
```

This loads content from **data.txt** into the element with ID content.

Benefits of AJAX

- Faster web applications
- No full page reload
- Better user experience

9. Write a program using jQuery to hide and show a paragraph when a button is clicked

Ans:

Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

```
<script>
```

```
$(document).ready(function(){
```

```
$("#hide").click(function(){
```

```
    $("p").hide();
```

```
});
```

```
$("#show").click(function(){
```

```
    $("p").show();
```

```
});
```

```
});
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<p>This is a paragraph.</p>
```

```
<button id="hide">Hide</button>
```

```
<button id="show">Show</button>
```

```
</body>
```

```
</html>
```

Output

- Clicking **Hide** hides the paragraph.
- Clicking **Show** displays the paragraph.

10. Write a jQuery program demonstrating append(), prepend(), and remove() methods

Ans:

Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

```

<script>
$(document).ready(function(){

$("#append").click(function(){
    $("#p").append(" Appended text.");
});

$("#prepend").click(function(){
    $("#p").prepend("Prepended text. ");
});

$("#remove").click(function(){
    $("#p").remove();
});

});
</script>

</head>

<body>

<p>This is a paragraph.</p>

<button id="append">Append</button>
<button id="prepend">Prepend</button>
<button id="remove">Remove</button>

</body>
</html>

```

Explanation

- **append()** adds text at the end of paragraph.
- **prepend()** adds text at the beginning.
- **remove()** deletes the paragraph element.